Chasing the Nines: Reliability Across States and Seeds

A Fixed-Grid Pendulum Case Study

Project 15

25 July 2026

Abstract

Reliable reinforcement learning must work across initial states and independent training seeds. We evaluate five trained actors on the same 2,501-state Pendulum grid. The selected reinforcement-learning-plus-supervision recipe combines 30,000 learner-controlled DAgger transitions, a 20,000-transition automatically prioritized DAgger follow-up, and a shared 50,000-transition FastSACN8 critic for conservative local Q-search. It succeeds near the reference on 12,496 of 12,505 seed-state trials, or 99.928%. The selected pure-RL pipeline uses 100,000-step SimbaV2 SAC actors with reflection and unanimous global Q-search. It reaches 11,832 trials, or 94.618%. Task-success rates are 93.858% and 92.499%; strict reference wins are 12.555% and 18.417%. The comparison is descriptive because supervision, discovery cost, and critic replication differ. Seed-resolved diagnostics weaken broad missing-coverage and widespread-dormancy explanations. They instead identify a selected pure-RL tail associated with compounding trajectory drift and imperfect critic ranking.

1 Question, evidence, and protocol

Chasing two kinds of nines. The project asks how to increase reliability within one trained model and across random training seeds. A high pooled score can conceal different failures in different seeds. Every mainline method is therefore evaluated on 61 angles in $[-\pi, \pi]$ and 41 angular velocities in $[-1, 1]$. Five actor seeds yield 12,505 deterministic 200-step rollouts in Gymnasium `Pendulum-v1`.

The observation, torque, and reward are

$$s = (\cos\theta, \sin\theta, \dot{\theta}), \quad a \in [-2, 2],$$

$$r_t = -(\theta_t^2 + 0.1\dot{\theta}_t^2 + 0.001a_t^2).$$

The scoring reference is the better stored return from a dynamic-programming policy and a controller, $R^\star(s) = \max(R_{\mathrm{DP}}(s), R_{\mathrm{ctrl}}(s))$. It is a strong comparator rather than a certificate of optimality.

For policy return $R_\pi(s)$, near-reference and strict-win indicators are

$$\mathcal{N}(s) = \mathbb{1}[R_\pi(s) \geq R^\star(s) - 5],$$
$$\mathcal{B}(s) = \mathbb{1}[R_\pi(s) > R^\star(s)].$$

Task success requires at least 80% of steps near upright and no not-upright streak longer than 50 steps. A step is near upright when $\cos\theta \geq 0.95$ and $|\dot{\theta}| \leq 1$.

Evidence selection. The two-repository census contains 404 standardized evaluations. Headline methods require the common grid, five actor seeds, no inference-time reference, no angle router, no policy-checkpoint mixture, and no more than 100,000 selected-route learning transitions. Automatic performance-based training-state selection and reference-free Q-search are allowed. Every displayed scorecard row contains five actors and 12,505 trials. Method and checkpoint selection was retrospective on the standardized grid campaign, with no independent held-out confirmation.

The mixed initializer used a manually biased supervised sampler: 20% of its 400,000 static labels came from $|\theta| \leq 35°$ and $|\dot{\theta}| \leq 1$. The result stays in the mainline under the final project decision, with this limitation disclosed. The 20,000-transition priority stage selects starts from measured performance rather than angle. Deployment uses no reference or angle-dependent rule.

Technical context and contribution. SAC trains a stochastic actor and twin critics from reward and entropy [1]. SimbaV2 contributes normalized residual networks for stable deep RL [7]. DAgger supplies reference labels on learner-visited states to address sequential distribution shift [3]. Our measured-regret priority stage is related to automatic curricula [4]. SACn contributes multi-step critic targets, while FastSACN keeps fewer horizons [8]. Critic action search is a small gated analogue of critic-guided action optimization [6]. Reflection and search are deployment projections rather than new training algorithms. Multi-seed evaluation is essential because aggregate means can conceal instability [9, 10].

This work contributes a systematic two-repository evidence census, combines the inherited components into two reference-free deployments, measures the paired contribution of each retained component, and diagnoses the remaining pure-RL tail. The analysis tests three explanations for that tail: missing experience, failed actor extraction, and critic-ranking error. It does not claim a new SAC or DAgger algorithm.

2 The two selected recipes

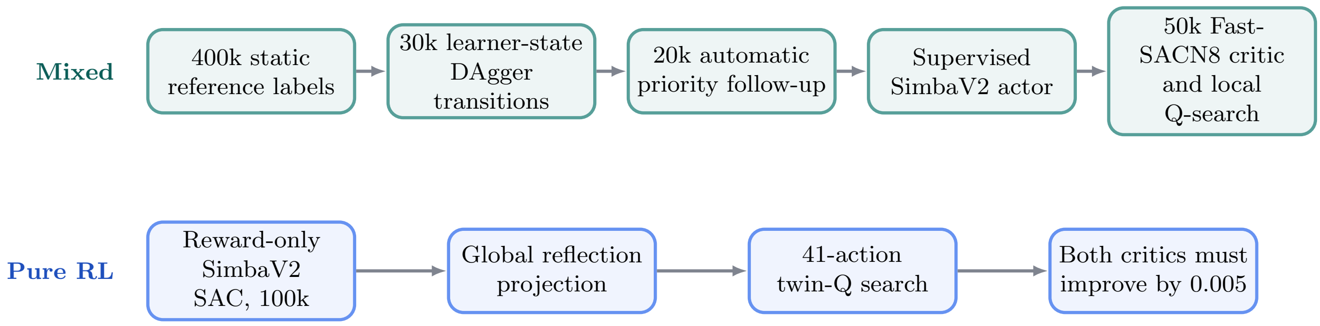

Figure 1: The mixed actor receives direct action labels during training and a reward-only critic corrects its action locally at deployment. Its separate reset-support and priority queries are detailed below. Pure RL learns from reward and uses symmetry plus a wider critic search. Both deployments are reference-free.

Mixed recipe in plain language. A width-64, two-block SimbaV2 actor first learns reference actions. Three learner-controlled DAgger rounds add the states that the current actor actually visits. A follow-up evaluates uniformly sampled candidate starts, ranks them by measured reference regret, and collects 20,000 transitions from the worst measured starts plus a uniform fraction. The chosen actor keeps the reference action as its supervised target. Tiny critic-based label shifts changed none of the requested classifications.

The supervised actor minimizes

$$\mathcal{L}_{\mathrm{BC}}(\theta) = \frac{1}{|\mathcal{D}|} \sum_{(s,a^\star)\in\mathcal{D}} \left(\frac{\pi_\theta(s) - a^\star}{2}\right)^2 .$$

At deployment, the actor proposes $a_0 = \pi_\theta(s)$. A shared reward-only FastSACN8 critic checks

$$\mathcal{A}_{\mathrm{local}} = \mathrm{clip}(a_0 + \{-0.10, -0.05, 0, 0.05, 0.10\}, -2, 2)\,.$$

The maximizing proposal replaces a_0 only when both critics prefer it. The critic target combines one-step and eight-step endpoints. With $\lambda = 0.5$, the normalized eight-step coefficient is

$$\frac{0.5^7}{1+0.5^7} = 0.775\%.$$

This critic is therefore dominated by its one-step endpoint.

Pure-RL recipe in plain language. Five independent SimbaV2 actor-critic systems train for 100,000 reward-only transitions. No reference action, reference return, demonstration, or manually selected training region enters learning. Each actor has width 32 and one residual block. Each twin categorical critic has width 64, two residual blocks, and 51 return atoms [2].

SAC minimizes $\mathbb{E}[\alpha \log \pi_\theta(a|s) - \min_i Q_i(s,a)]$ for the actor and fits each critic to $r + \gamma(\min_i \bar{Q}_i(s',a') - \alpha \log \pi(a'|s'))$. Seeds are 0–4. Adam uses batch 256, replay capacity 100k, $\gamma = .99$, $\tau = .005$, and actor/critic rates decaying from 10^{-4} to 5×10^{-5}; critics update once per transition and the actor every second update.

The deployment projection uses Pendulum sign symmetry:

$$M(s) = (\cos\theta, -\sin\theta, -\dot{\theta}),$$
$$a_{\mathrm{sym}}(s) = \frac{\pi_\theta(s) - \pi_\theta(M(s))}{2}.$$

The critics then evaluate 41 torques in $[-2, 2]$. Proposal $a_q = \arg\max_a \min_i Q_i(s,a)$ is accepted only if

$$Q_i(s,a_q) - Q_i(s,a_{\mathrm{sym}}) > 0.005 \quad \text{for } i \in \{1,2\}.$$

Reflection is one global mathematical rule. Q-search uses learned reward critics and never reads the reference.

Resource accounting. The selected mixed route assigns 30k initial DAgger transitions, 20k priority DAgger transitions, and one 50k shared critic, giving 100k learning transitions. Its 690k reference queries comprise 400k initializer labels, 30k learner-state labels, 240k reset-support labels, and 20k priority labels. It also uses 800k candidate-rollout interactions to rank starts. Across five actors, the actual mixed study contains 300k learning transitions because the 50k critic is shared. Pure RL uses 100k learning transitions per seed and 500k across five complete actor-critic runs. The result compares selected pipelines rather than equal total compute.

Table 1: Selected-route and full-study resource ledgers.

Pipeline	Learn/route	Learn/full	Discover/full	Labels/full
Auto-priority mixed	100k	300k	4.00m	3.45m
Pure RL	100k	500k	0	0

3 Reliability and component results

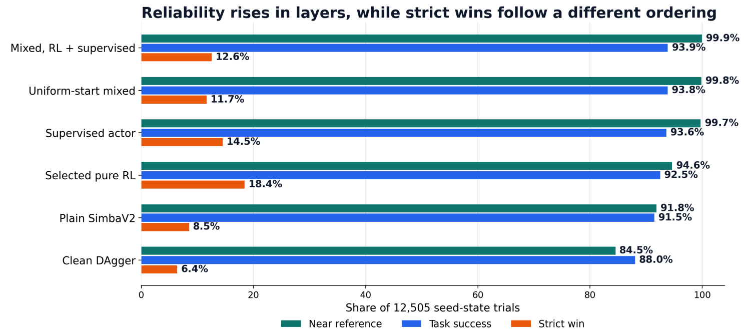

Figure 2: Scorecard on the same 12,505 seed-state trials. Every row uses five actors; the mixed rows share one selected critic.

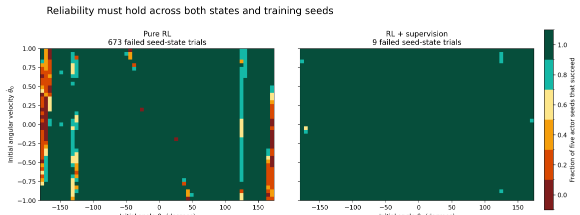

Figure 3: Near-reference success fraction across five actor seeds at every initial state.

The mixed method controls the reliability tail. The selected mixed pipeline reaches 12,496 of 12,505 near-reference trials, compared with 11,832 for pure RL. It also records 11,737 task successes versus 11,567. The selected pure pipeline records 2,303 strict wins versus 1,570 for the mixed pipeline. This descriptive difference shows that reliability and exceeding the stored comparator induce different pipeline rankings.

The gap is seed-dependent. Mixed near-reference counts range from 2,494 to 2,501 of 2,501 states per seed. Pure counts range from 2,334 to 2,407. The mixed pipeline succeeds for all five seeds on 2,493 cells, or 99.680%, and every cell succeeds for at least one seed. Pure RL succeeds for all five on 2,255 cells, fails for all five on 39, and varies by seed on the remaining 207 cells.

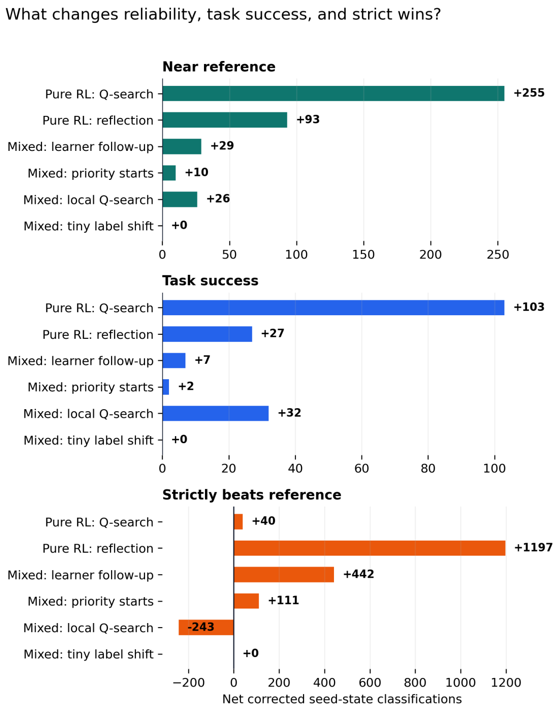

Figure 4: Matched fixed-minus-broken classification counts.

Mixed ablations. The 20k learner follow-up adds a net 29 near-reference, seven task-success, and 442 strict-win classifications. Automatic priority adds ten, two, and 111 relative to uniform starts. Local FastSACN8 Q-search adds 26 near-reference and 32 task-success classifications, but loses 243 strict wins. Tiny critic shifts to supervised labels change zero requested classifications.

Pure ablations. Reflection adds 93 near-reference and 27 task-success classifications, together with 1,197 strict wins. Global unanimous Q-search adds another 255 near-reference, 103 task-success, and 40 strict-win classifications. These paired deployment interventions show that the actor leaves useful critic and symmetry information unexpressed.

4 Why pure reinforcement learning leaves a larger tail

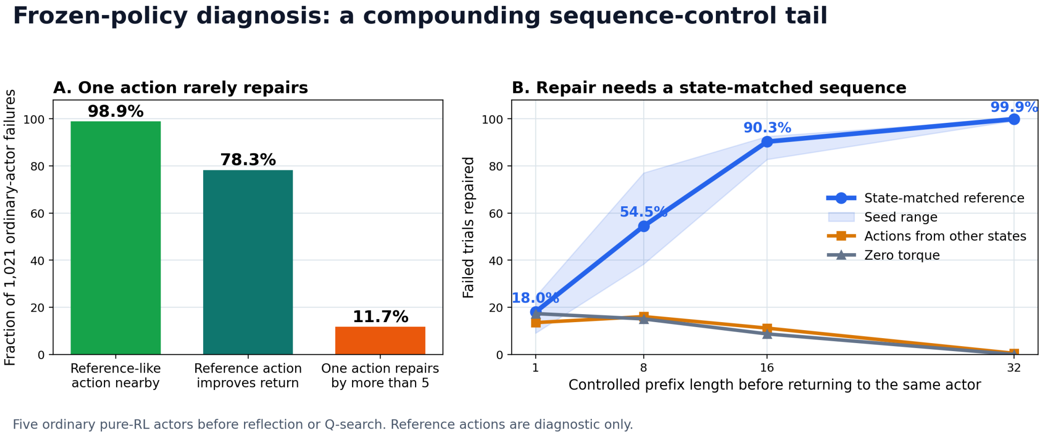

Figure 5: Frozen-policy interventions on five pure-RL actors before reflection and Q-search. Reference actions are diagnostic only.

Broad reset coverage is not the separator. Among 321 cells containing successful and failed pure seeds, the failed seed has a farther nearest training reset in 51.1% of cells, with sign-test $p = 0.738$. Its 256-neighbor replay radius is larger in only 24.9%. A reference-like action occurs among nearby replay actions for 1,010 of 1,021 ordinary-actor failures, or 98.9%. In 2,500 exact-start comparisons, a stored training trajectory beats the terminal actor by more than five return units in only 1.16%. These tests weaken explanations based on coarse reset absence or wholesale failure to reuse complete trajectories.

Reliable repair is sequential. A reference first action produces any positive return change in 78.3% of ordinary-actor failures, but a practically decisive realized gain above five occurs in only 11.7%. A separate state-matched prefix experiment asks whether the whole trajectory becomes near-reference after control returns to the actor. It repairs 18.0%, 54.5%, 90.3%, and 99.9% after 1, 8, 16, and 32 reference-controlled steps. At 32 steps, shuffled reference actions repair 0.5% and zero torque repairs none. The result supports a sustained state-dependent correction rather than a generic interruption.

The critic is useful and imperfect. On 2,259 pure first-disagreement failures, the critic-ranking diagnostic records 0.782 precision, 0.703 recall, 0.623 balanced accuracy, and 0.655 ROC AUC for helpful local proposals. Global Q-search therefore has useful net value while retaining ranking errors in the tail.

Optimization and representation checks. Completed SimbaV2 arms log zero dormant units at the registered threshold. This rules out widespread measured dormancy rather than all plasticity or optimization problems [14, 15, 11]. Effective-rank and local parameter-plane diagnostics vary across checkpoints but do not currently separate the selected categories strongly enough for a causal claim. They stay in the annex.

What the FastSACN8 combination changed. With the same 100k budget, SimbaV2 architecture, and deployment, but a changed target and critic dose, FastSACN8 plus critic UTD2 lowers ordinary-actor near-reference reliability from 91.864% to 89.937% on 27,765 locked off-grid trials. The same reflection and Q-search instead raise 94.489% to 95.909%; Q-search adds 3.584 points after reflection versus 1.080. Task success loses 25 trials, so the registered gate fails and the authority grid is not queried. Frozen geometry shows higher saturation and lower action sensitivity in all five paired seeds. The combination strengthens critic-guided repair without improving raw actor reliability.

Best-supported account. Pure RL visits relevant regions, stores nearby corrective actions, and often learns critics with useful local rankings. Its remaining errors concentrate in a seed-dependent state tail where one action rarely repairs the trajectory. The selected mixed pipeline supplies direct corrective labels on learner states and automatically concentrates its follow-up on measured failures. The observed gap is therefore consistent with more reliable sequence control under supervision, combined with imperfect actor-critic transfer in pure RL.

Limits and conclusion. The pipelines differ in labels, discovery interactions, update histories, critic replication, and sampler design. The fixed grid gives exact benchmark counts rather than independent population trials. Frozen-policy diagnostics localize failure mechanisms but cannot identify the optimizer update that created them. Within those limits, the selected mixed method reaches 99.928% near-reference reliability with nine failures, compared with 673 for the selected pure-RL deployment at 94.618%. Pure RL produces more strict wins. Chasing the next nine requires better actor-side sequence control; stronger critic-guided deployment did not close the reliability gap.

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A Evidence and diagnostics annex

A.1 Evidence tiers and provenance

Tier A evidence uses standardized five-seed rollout tables on the registered 61 by 41 grid. Tier B evidence uses frozen checkpoints and registered diagnostic state sets after method selection. Tier C evidence contains short screens, local parameter perturbations, and one-seed probes. Only Tier A results determine the headline methods. Tier B evidence supports mechanism interpretation. Tier C evidence is exploratory.

The selected mixed actor family is the registered no-shift automatic-priority DAgger follow-up. The selected pure deployment is the registered SimbaV2 symmetric-actor Q41 unanimous-search evaluation. Both contain five actor indices and 12,505 fixed-grid rollouts.

The mixed initializer command used 400,000 static labels, three learner-controlled DAgger rounds, and a 20% near-upright sampler within 35 degrees. The follow-up used 240,000 reset-support labels plus 20,000 automatically prioritized learner states. This provenance is reported as a sampler limitation. The deployment actor and local Q-search do not read the reference or use an angle rule.

Demonstration-augmented policy gradients provide another route for combining imitation and reward optimization [5]. The selected mixed route instead keeps the supervised actor target separate from the reward-trained deployment critic.

A.2 Systematic two-repository census

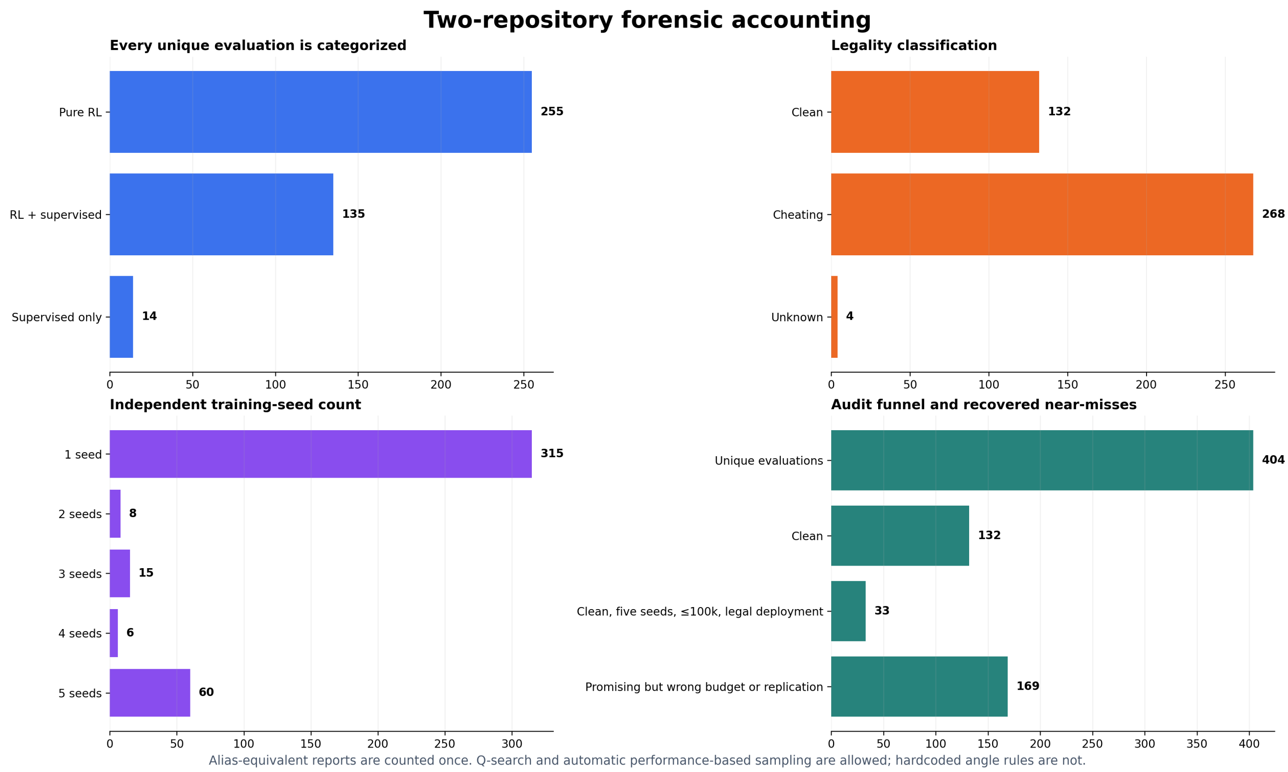

Figure 6: Forensic inventory of standardized evaluations in Project 15 and Project 15-sac-n-experiments. Counts describe the crawl before the final project decision retained the accepted mixed result with a sampler footnote. The result tables in the main report follow the final decision.

The inventory is useful for identifying missing seeds, aliases, deployment variants, and unreported experiment families. Its earlier binary training-sampler classification is not used to replace the accepted mainline result.

A.3 FastSACN8 target weight and learning curves

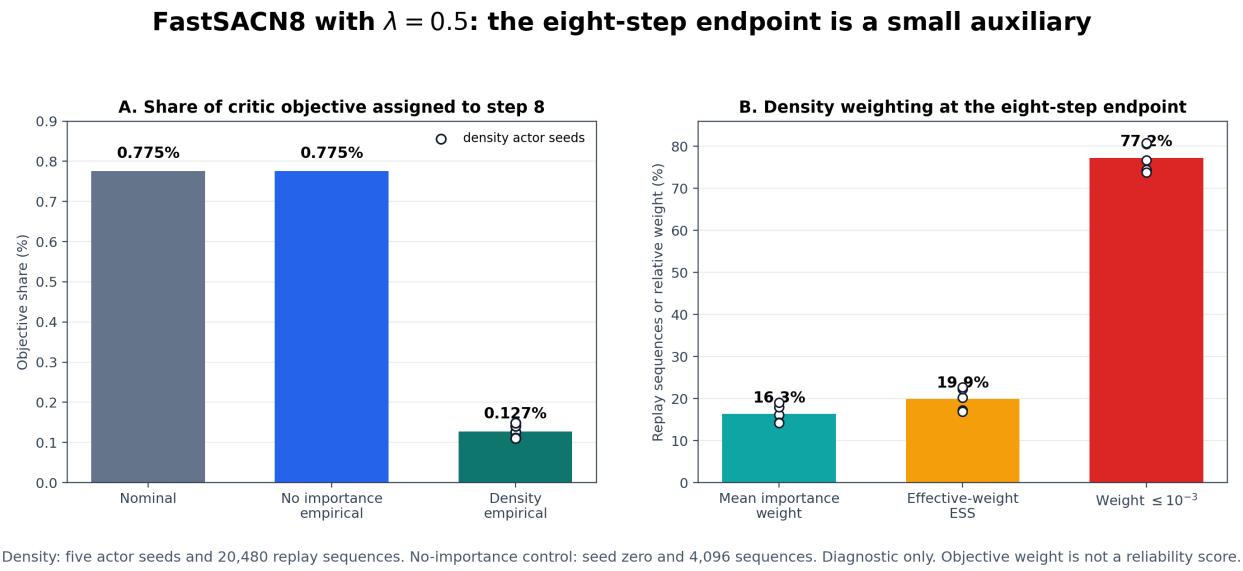

Figure 7: Normalized horizon weights for FastSACN target variants. The $\lambda = 0.5$ FastSACN8 recipe assigns 0.775% of normalized target mass to the eight-step endpoint.

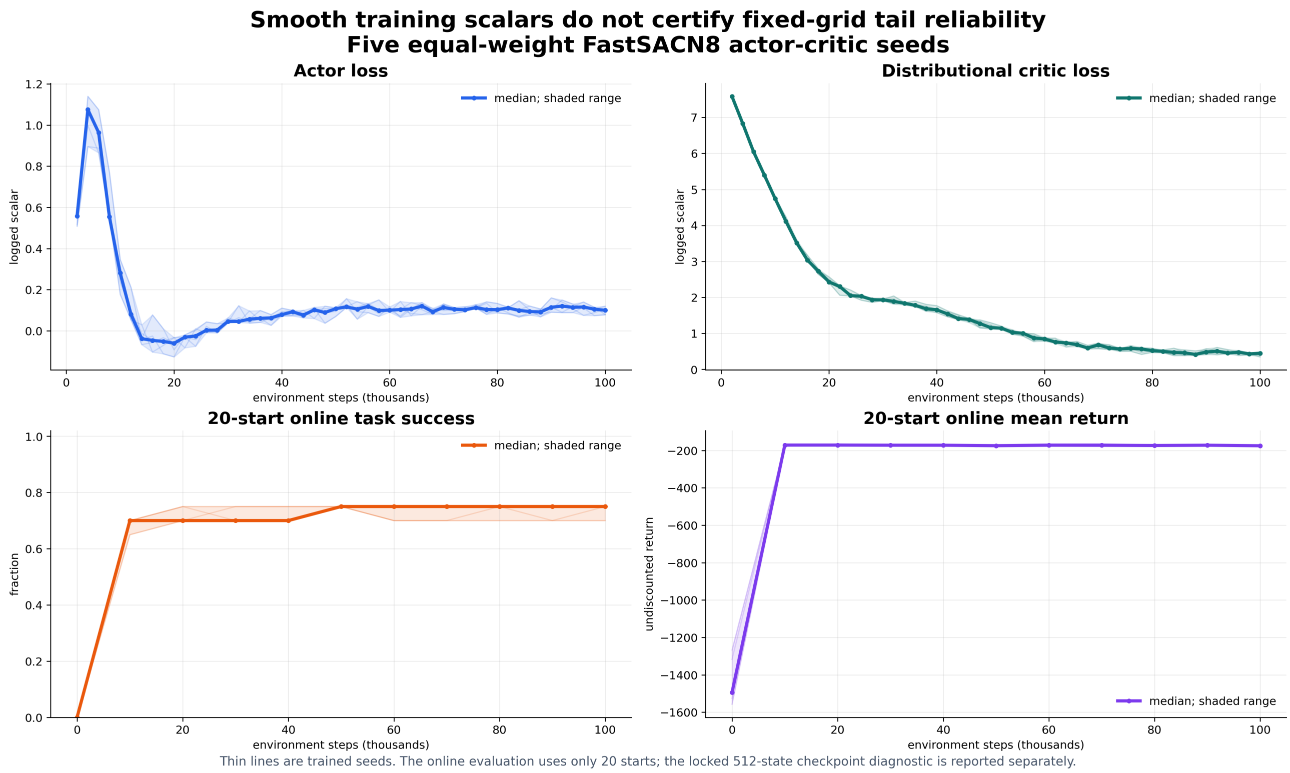

Figure 8: Stored loss and evaluation traces for the earlier FastSACN8 family. Loss scales differ across objectives, so the curves describe optimization history rather than provide a direct cross-objective ranking.

The historical FastSACN8 result changes training length, critic update dose, and deployment relative to the selected one-step pure method. The completed P7 combination keeps the 100k budget, SimbaV2 architecture, and deployment fixed while changing both the target family and critic update dose. On 27,765 locked off-grid trials per arm, its selected deployment improves near-reference reliability from 94.489% to 95.909% and strict wins from 17.792% to 21.930%, while task success decreases from 93.038% to 92.948%. The registered no-regression gate therefore blocks authority-grid evaluation. This comparison tests the combined recipe rather than the target family alone.

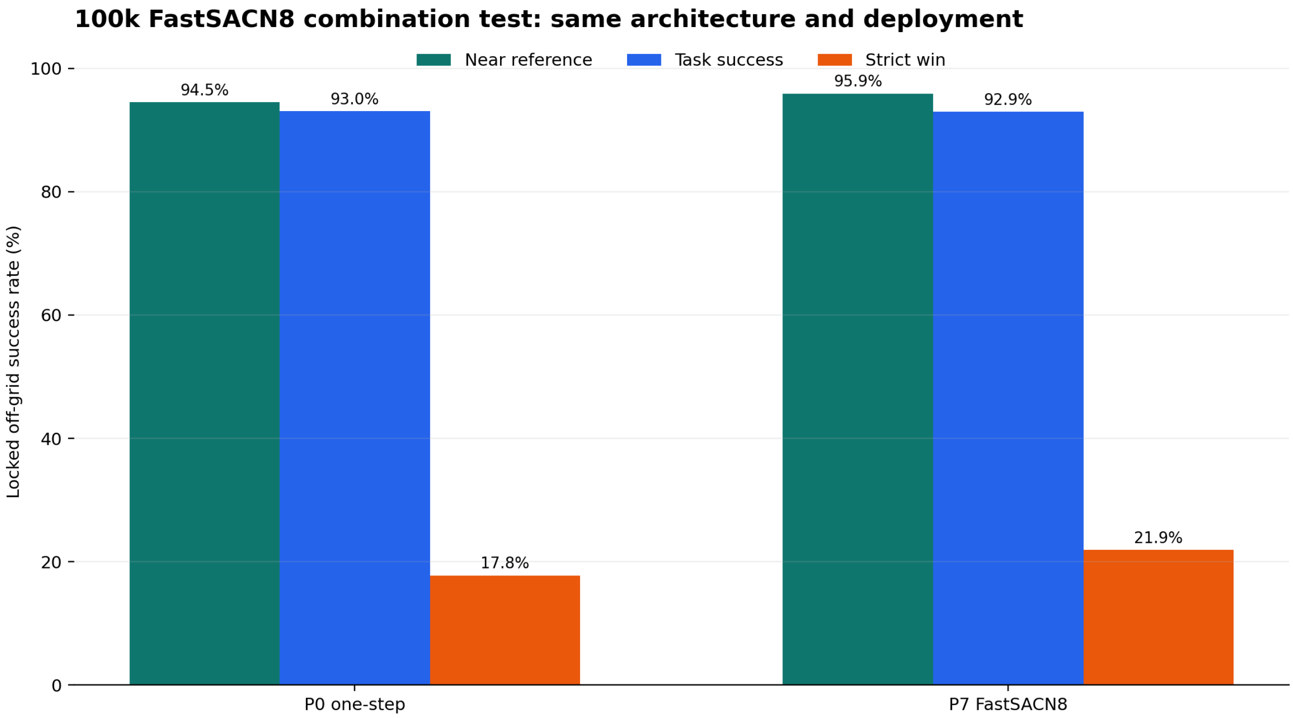

Figure 9: Five-seed locked off-grid comparison under the same reflection and Q-search deployment. P7 changes both the FastSACN8 target and critic update dose. The authority grid was not queried.

The P8 seed-zero actor-UTD2 screen improves strict wins and mean return relative to P7 seed zero, but lowers near-reference reliability from 96.668% to 96.128% and task success from 92.527% to 92.112%. It fails its registered seed-zero gate and receives no four-seed promotion.

A.4 Actor geometry and saturation

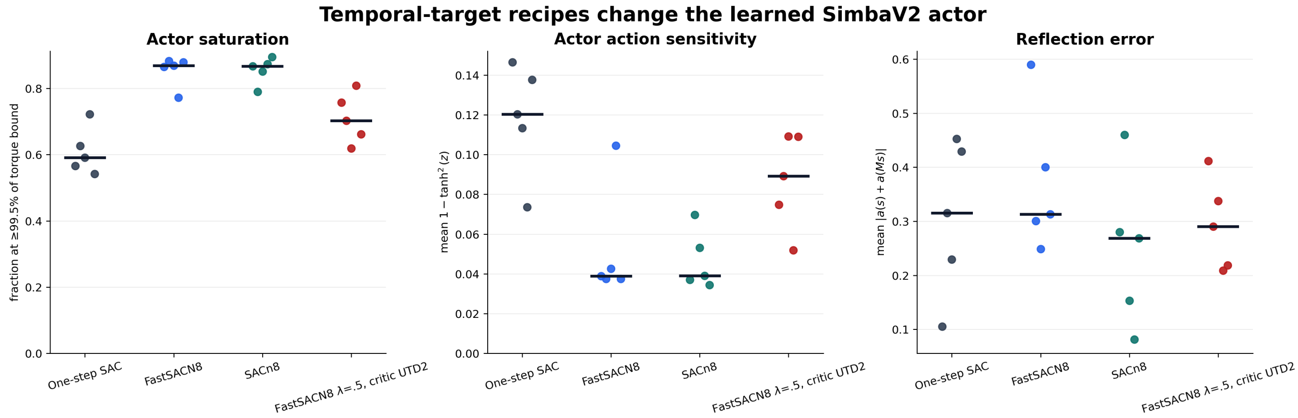

Figure 10: Checkpoint-resolved actor saturation, action sensitivity, and reflection error for completed temporal-target arms. These actor-geometry measures are descriptive and do not identify the effect of critic update dose.

Relative to the five one-step actors, the P7 combination raises the median saturation fraction from 59.1% to 70.3% and lowers the median tanh action-sensitivity statistic from 0.120 to 0.089. Both paired changes have the same direction in all five indexed seeds. Its ordinary actor is less reliable, while its critic-guided deployment is more reliable. This pattern associates the gain with deployment repair rather than a better raw actor, but the target and critic dose change together.

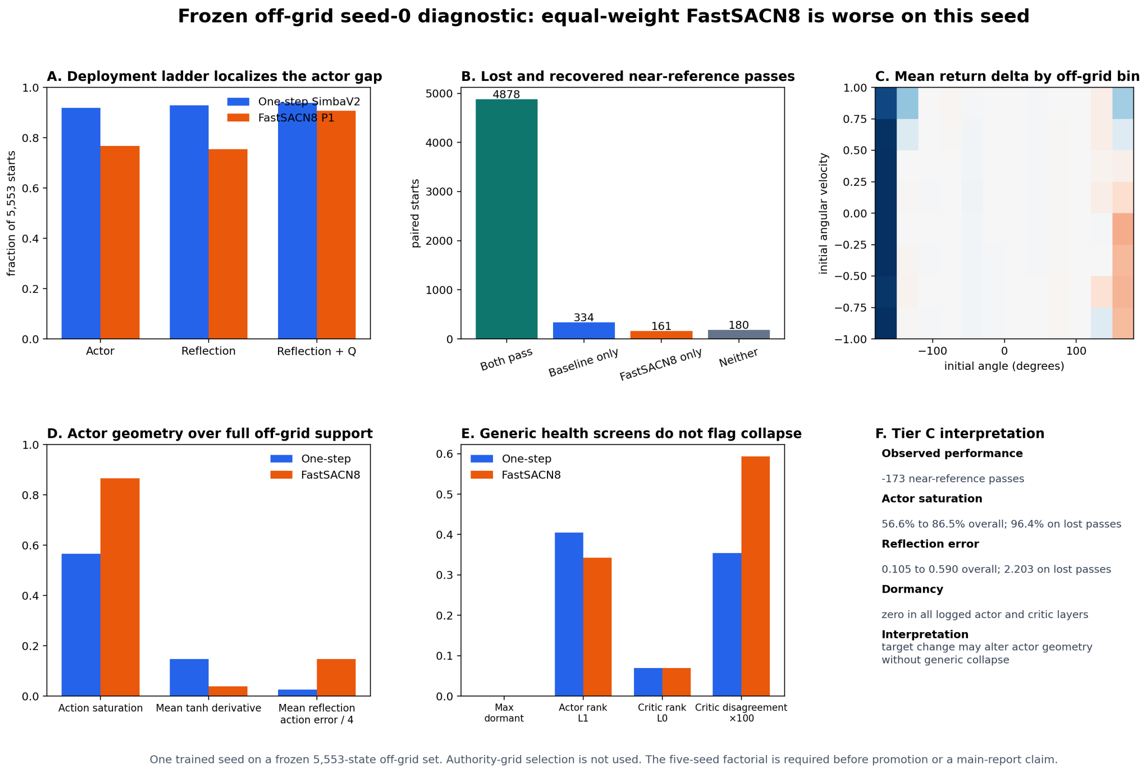

Figure 11: Seed-0 FastSACN8 actor geometry probe. This one-seed analysis is exploratory and cannot become a headline result.

Pure-actor failures are strongly associated with saturated actions, but the reference action is also saturated on nearly all of those states. Saturation therefore identifies a difficult swing-up regime rather than proving that the actor chose the wrong direction. The first-action intervention separates a smaller opposite-sign error group from a larger same-sign timing group.

A.5 Dormancy, effective rank, and telemetry

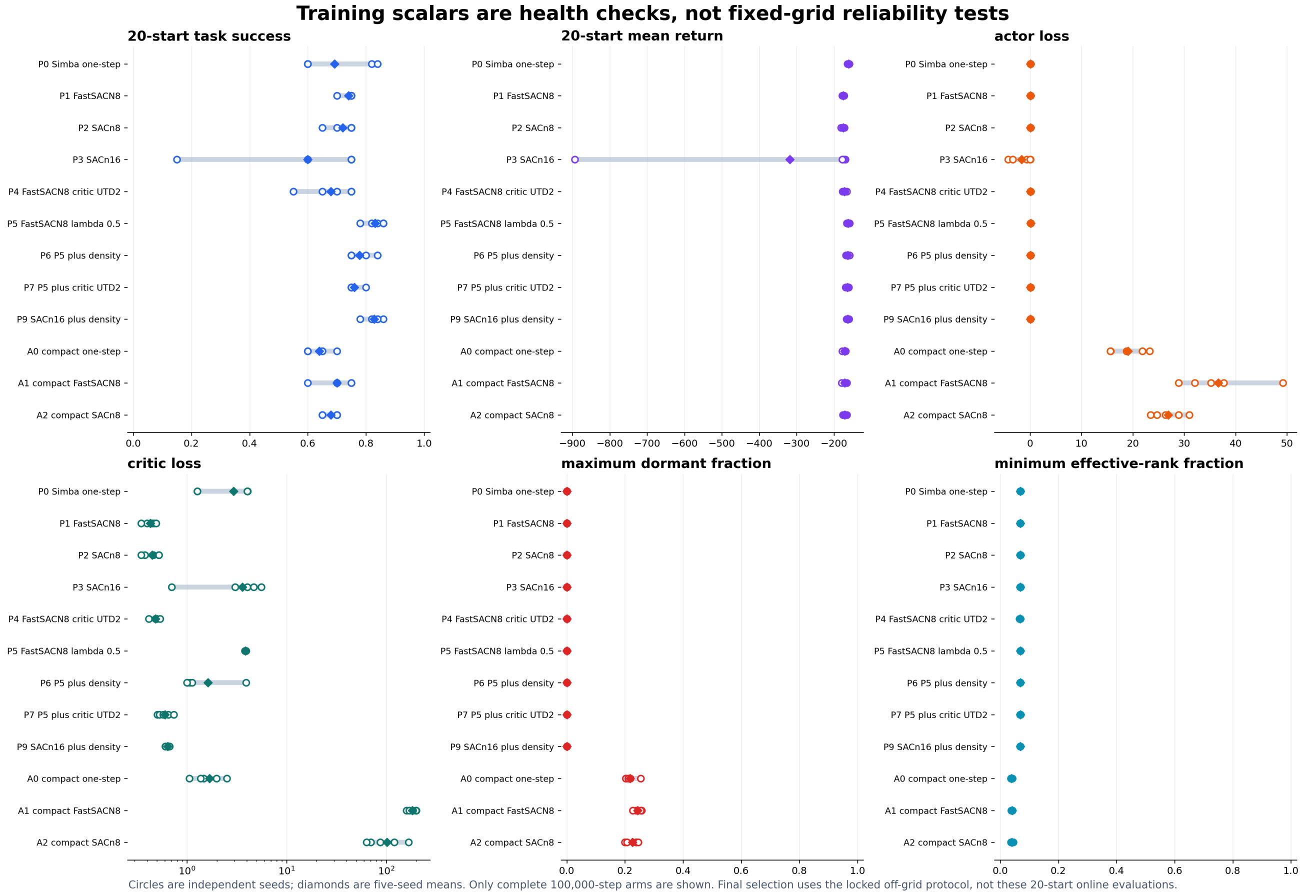

Figure 12: Seed ranges for online evaluation, actor loss, critic loss, maximum measured dormancy, and minimum effective rank across completed five-seed pure-RL arms.

The separation between generated experience and the policy extracted from it follows exploration-versus-optimization diagnostics [13]. The registered threshold finds zero dormant units in the completed SimbaV2 arms. This is a negative result. It rules out widespread measured dormancy as the explanation for the selected reliability gap. It does not establish healthy plasticity, well-conditioned gradients, or good critic calibration. Effective rank, plasticity, and policy churn are narrower complementary checks [16, 12, 17]. Loss magnitudes also differ across target objectives and should not be ranked without normalization.

A.6 Local parameter landscapes

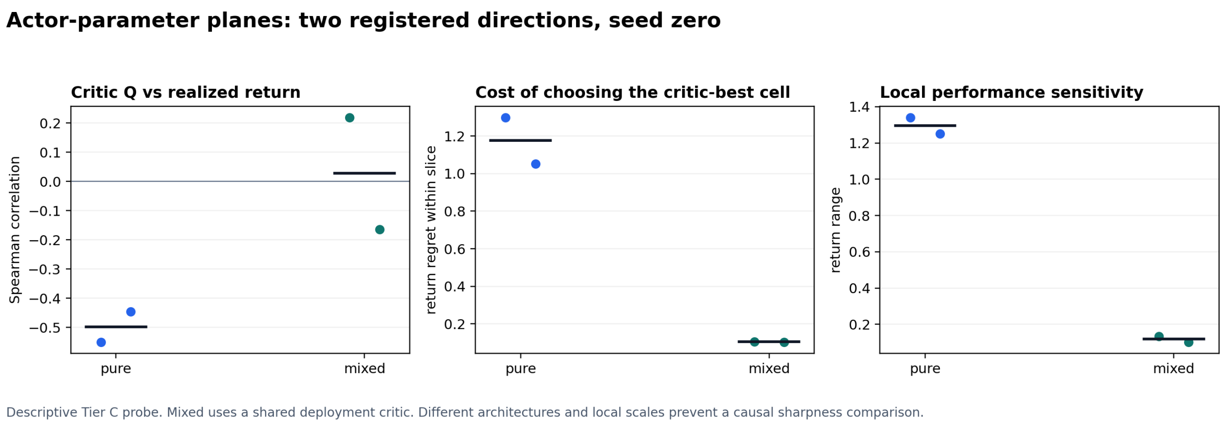

Figure 13: Local parameter perturbation planes around selected frozen checkpoints. The geometry depends on perturbation directions, parameter scaling, and the fixed diagnostic batch.

The planes show local sharpness and coupling for the chosen directions. They do not provide a global loss landscape or a causal explanation. Their role is to check whether a method ends in an obviously unstable local neighborhood. The current result is exploratory because the selected categories use different objectives and datasets.

A.7 First disagreement and critic ranking

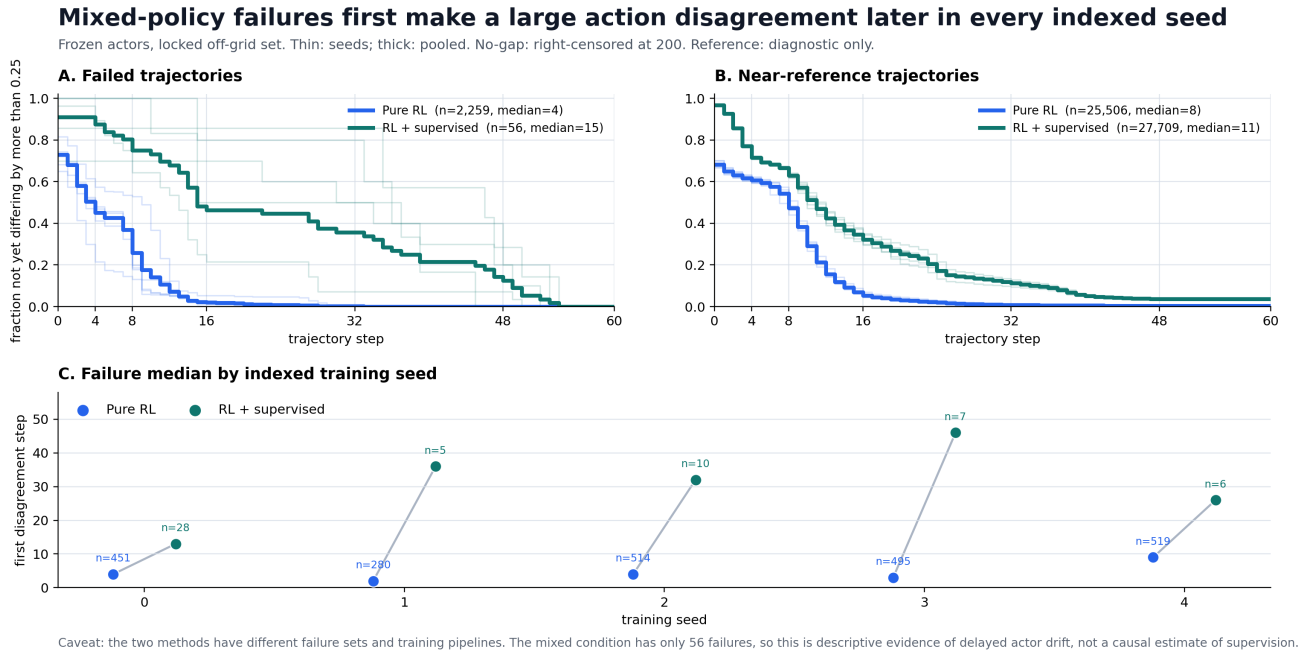

Figure 14: Step of first large actor-reference action disagreement on a locked off-grid set. Thin curves show indexed actor seeds and thick curves pool failure rows. Different failure sets make the category comparison descriptive.

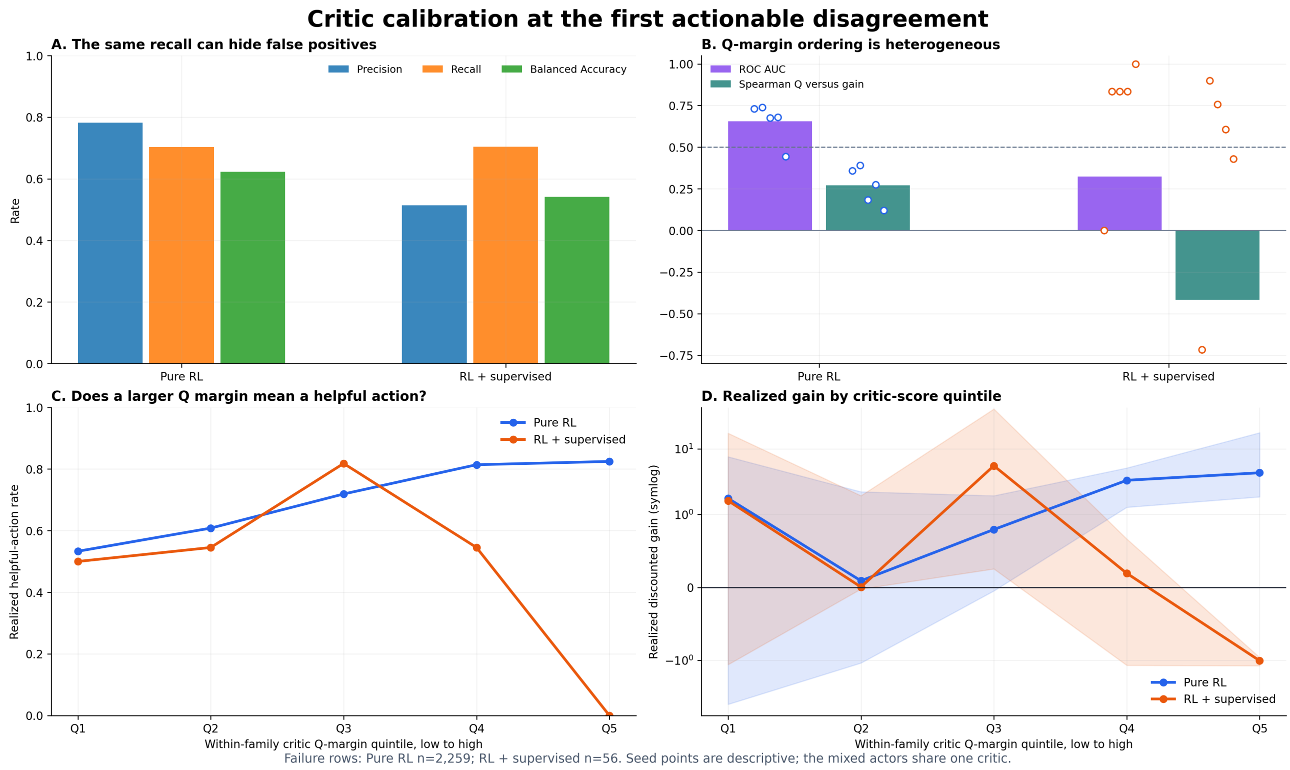

Figure 15: Critic proposal calibration at first-disagreement states. Precision, recall, balanced accuracy, AUC, and Q-margin bins separate useful local ranking from harmful proposals.

For pure RL, the pooled first-disagreement diagnostic contains 2,259 failure rows. Helpful-proposal precision is 0.782, recall is 0.703, balanced accuracy is 0.623, and ROC AUC is 0.655. The critic contains useful information but does not reliably order every hard-state proposal. This result is consistent with the positive net gain and nonzero break count of global Q-search.

A.8 Seed-resolved timing

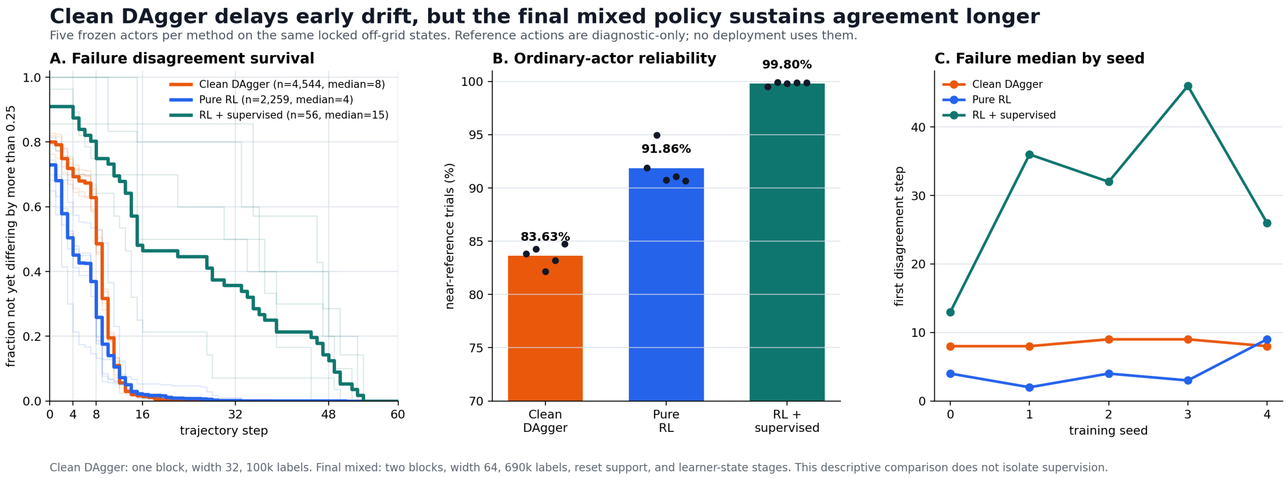

Figure 16: Seed-resolved trajectory timing for pure RL, the selected mixed pipeline, and DAgger timing controls. Counts are displayed because the failure populations differ substantially.

The mixed failures diverge later than pure failures on the selected frozen sets. This observation is consistent with learner-state supervision reducing compounding drift. It is not a causal estimate because the pipelines differ in labels, initialization, interaction counts, critics, and selection history.

A.9 Closed-loop sensitivity

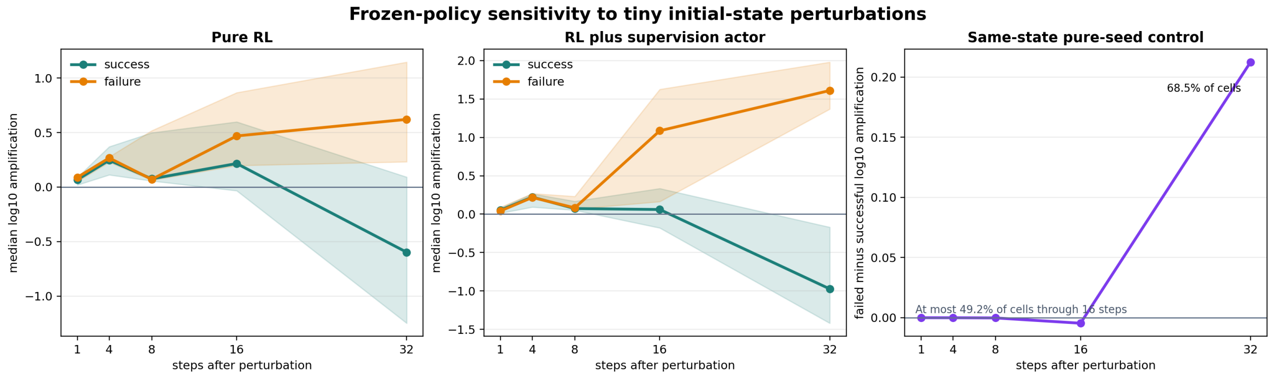

Figure 17: Closed-loop response to small state perturbations on selected frozen policies. The diagnostic measures trajectory amplification after selection.

Closed-loop amplification tests whether small early deviations grow into a reliability failure under the same frozen actor. It complements the action-ranking diagnostic: a good local action is insufficient when later policy actions amplify residual state error.

A.10 Reference-prefix specificity

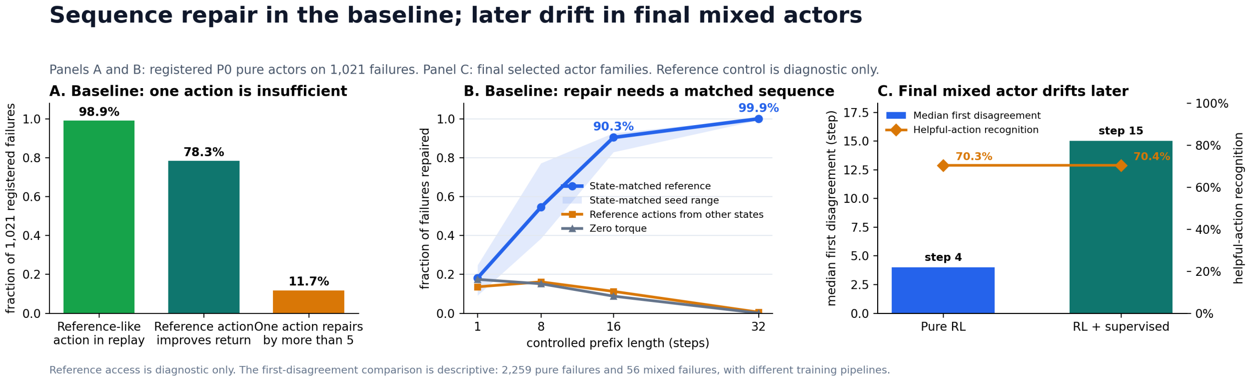

Figure 18: Repair rate as a state-matched reference policy controls the first 1, 8, 16, or 32 steps and then returns control to the same frozen actor.

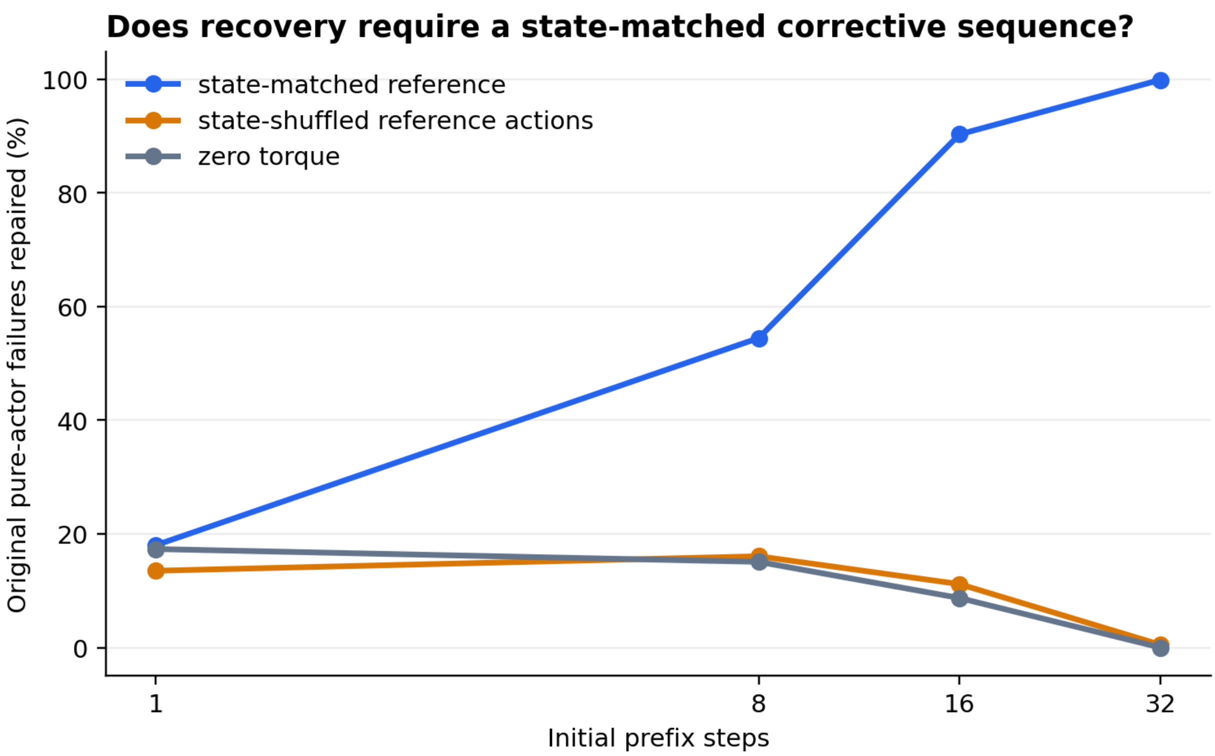

Figure 19: Specificity controls at 1, 8, 16, and 32 steps. Shuffled reference actions and zero torque do not reproduce the state-matched repair at longer prefixes.

The intervention demonstrates sufficiency on the tested frozen failures. A sustained state-matched prefix repairs nearly every trial, while generic interruptions do not. The reference is diagnostic only and is unavailable at deployment. This result does not show which training update would teach the same corrective sequence.

A.11 Interpretation boundary

The combined diagnostics reject several simple explanations at their measured resolution:

- Failed seeds do not systematically have farther nearest training resets.
- Reference-like actions are usually present in nearby replay.
- Complete stored trajectories rarely outperform the terminal actor by more than five return units.
- Widespread dormant units are absent at the registered threshold.

They support a narrower account:

- Pure failures often begin with early actor-reference disagreement.
- One local action rarely removes the complete return deficit.
- State-matched corrective sequences are sufficient on almost all tested failures.
- The learned critic ranks helpful proposals better than chance but makes consequential tail errors.

These conclusions apply to the selected Pendulum checkpoints and diagnostic state sets. Broader claims require other environments, matched starts and budgets, independent critics, and a uniform initializer.